UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION 1 (5 marks)

Draw the Von Neumann architecture and give appropriate example for each component.

QUESTION 2 (5 marks)

Identify the correct generation based on the following descriptions.

a) The production of personal computer marks the main feature of this generation. The price of computer during this period is the lowest compared to the other generation. It provides fast processing ability with less power consumption.

b) Machines that emulate human intelligence which can process and respond to natural language is developed in this generation.

c) In this generation, air condition is no longer required because the heat produced by the machine is reduced. However, fan is needed for heat discharging.

d) Integrated circuit comprises of a number of transistors, resistors and capacitors are invented by Jack Kilby in this generation.

e) Assembly language is introduced during this generation to replace machine language for more reliable and improved performance.

QUESTION 3 (5 marks)

Solve the following equations in binary number bases. Show your workflow.

a) 1.0000 10012-0.0011 OOH2

b) O.FA3i6-0.651s

QUESTION 4 (5 marks)

The X machine was the first automatic electronic digital computer that uses binary system. It has 30 arithmetic/logic unit that can add and subtract, a rotating drum memory that can held thirty binary numbers of fifty digits each and input/output components. Figure 1 shows the architecture of X machine.

a) What is X machine?

b) Name the inventor of X machine.

c) Based on Figure 1, identify the input and output components for X machine.

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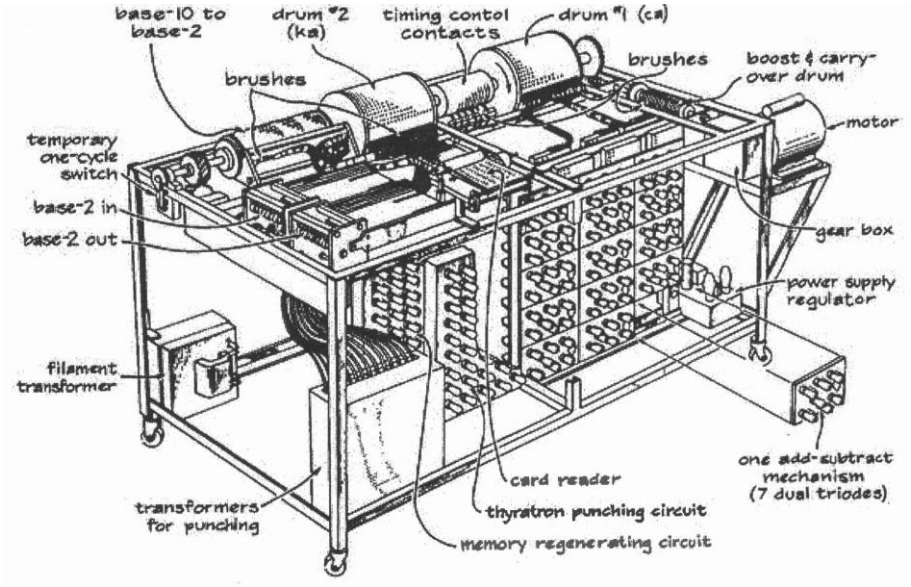


Figure 1: Machine X

QUESTION 5 (10 marks)

a) Discuss THREE (3) main key factors for memory selection.

b) List TWO (2) examples for internal cabling memory and TWO (2) examples of system memory.

QUESTION 6 (10 marks)

a) Provide one example for each classification of multiple processor organization.

b) Draw a simplified computer architecture of the following:
Symmetric Multiprocessors
ii. Clusters
iii. Non Uniform Memory Access (NUMA)

QUESTION 7 (10 marks)

a) When is an I/O module installed?

b) What will happen if the I/O module is not installed?

c) Explain ONE (1) category of I/O module

d) What does it mean by 'hot connection'?

e) List THREE (3) common interfaces to external devices.

QUESTION 8 (10 marks)

a) State TWO (2) different types of bus.

b) The size of a word is 32 bits and the size of the block is 256 bits. Determine of the following:

i. The byte for word

ii. The byte for block,

iii. Size of the register,

iv. Size of the data bus.

QUESTION 9 (10 marks)

a) Describe TWO (2) types of interrupt.

b) Briefly explain THREE (3) interfaces to external devices.

QUESTION 10 (10 marks)

Question 9 is based on Figure 2.

a) Identify addressing mode in Figure 2.

b) Which register will be involved in Figure 2 addressing mode?

c) If register in Figure 2 is replaced with memory, what is the new addressing mode?

d) Addressing mode in (a) is using register and addressing mode in (c) is using memory. Discuss which addressing mode will be executed faster with appropriate justification.

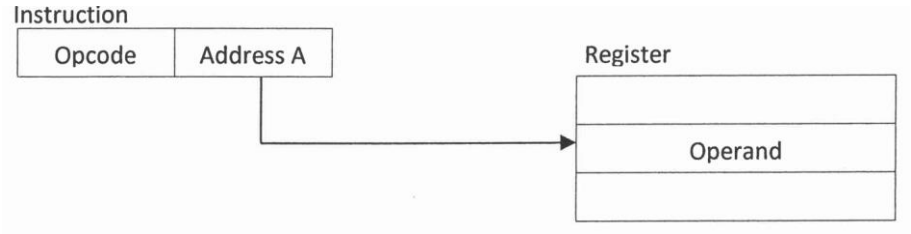


Figure 2: Addressing mode type x.

PART B (20 MARKS)

Answer the question in this part.

QUESTION 11 (20 marks)

Question 1 is based on Figure 3 and Table 1

Write an appropriate program using conect assembly language notations to compute the equation shown in Figure 3 using instructions for zero-address, one-address, and three-address. The available instructions are provided in Table 1. The registers labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register z.

$$z = A + C^2 / B * D - 3A$$

Figure 3

Table 1: lt/achine instruction

Zero-address	One-address	Three-address
PUSH p	LOAD p	ADD (p,q,r)
POP p	STOR p	SUB (p,q,r)
ADD	ADD p	MUL (p,q,r)
SUB	SUB p	DIV (p,q,r)
MUL	MUL p	
DIV	DIV p	